Problem 8

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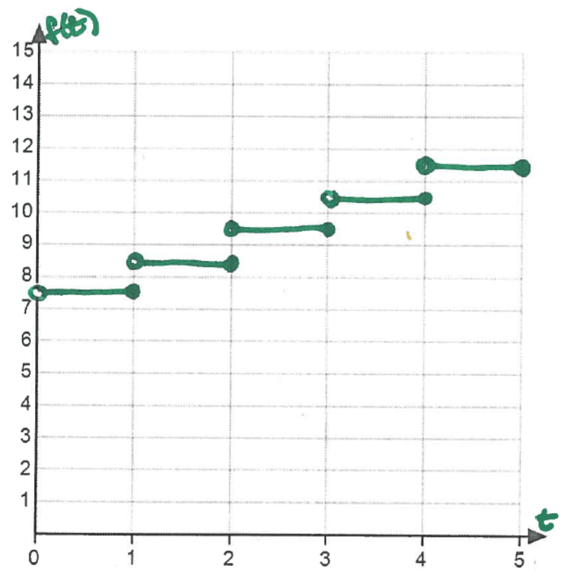
Problem 1 *Suppose a taxi ride costs* \$7.50 *for the first mile (or any part of the first mile), plus an additional* \$1.00 *for each additional mile (or any part of a mile).*

(a) *Graph the function $c = f(t)$ that gives the cost of a taxi ride for t miles, for $0 \leq t \leq 5$.*

Explanation.



(b) *Evaluate* $\lim\limits_{t\to 2.9} f(t)$

Explanation. $\lim\limits_{t\to 2.9} f(t) = 9.5$

(c) *Evaluate* $\lim\limits_{t\to 3^-} f(t)$ *and* $\lim\limits_{t\to 3^+} f(t)$

Explanation. $\lim\limits_{t\to 3^-} f(t) = 9.5$ and $\lim\limits_{t\to 3^+} f(t) = 10.5$

(d) *Interpret the meaning of the limits in part (c).*

Explanation. As the number of miles the taxi drives approaches 3, the cost of the taxi ride is \$9.50. If one drives for a bit more than 3 miles, the cost is \$10.50.

(e) *On what intervals is the function c continuous? Explain.*

Explanation. $(0, 1], (1, 2], (2, 3], (3, 4], (4, 5]$